Alex I.

SOFTWARE ENGINEER

Education
Computer Science and Software Engineering

Language proficiency
English – B2
Polish – B1
Swedish – A2

Domains
Telecommunications
Library Systems
Oil and Gas Industry
UAV Systems
Aerospace Systems

Software engineer with 13+ years of commercial experience.
I possess a strong background in both hardware and software development. My expertise mostly revolves around C/C++ software development, where I appreciate the complexity and versatility of the language, and MATLAB, where I have extensive experience in developing algorithms and applications for image analysis. I am eager to work on projects that prioritize performance and innovative features and enthusiastic about collaborating on challenging projects that allow me to leverage the full capabilities of these technologies and contribute to cutting-edge solutions.

Programming Languages
C++, C, C#, Java, JavaScript, Ruby, MATLAB, Scala, HTML.

Operating Systems
Windows, Embedded Linux.

Tools
Yocto Project, Qt Designer, Qt Designer .ui forms, System Tool Kit, systemd, CMake, qmake, gcc, GDB, Valgrind, Qt Test, Visual Studio, Qt Creator, Resharper C++.

Frameworks & Libraries
GStreamer, Qt, Qt Widgets, QSS, Ruby on Rails, .NET, GTest, Node.js, WinForms, FLTK, plog, PCL, V4L2, jQuery, Open3D, Qt DBus.

Multimedia
WebRTC, Qt Multimedia, Linphone-SDK, liblinphone, mediasoup, libmediasoupclient.

Databases
MySQL, PostgreSQL, SQLite, Microsoft SQL Server, MS Access.

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Network
SDP, Nginx, SIP, RTP, Qt WebSockets, SNMP, DHCP, TFTP.

Cloud
AWS.

Embedded
BitBake, NXP i.MX.

Development Processes
Jira, Asana.

Source control systems
Git, GitLab, Bitbucket.

Architecture
Qt Model/View.

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Projects

CLOUD NATIVE ENCODING PLATFORM

A distributed cloud-native microservices platform for real-time video encoding and delivery, designed to support 100K+ concurrent users across three availability zones. The platform handles both live and video-on-demand content with low latency and high reliability.

The codebase of the project uses a hybrid approach mixing C++ for performance critical services and Java/Kotlin for development speed. All services are cloud-agnostic allowing for even hybrid setups using both multiple clouds and on-premise infrastructure.

Project roles
Streaming Engineer/Architect

Period
04.2025 – till now

Responsibilities

- Replaced legacy JSON-RPC IPC with gRPC/Protobuf, reducing serialization overhead by ~40% and request size by ~60%;
- Analyzed and optimized cost/upload speed ratio for cloud client connectors to achieve the best fit based on customer preferences;
- Improved predictable segment upload SLA from 90% to 99.9% for cloud storages;
- Architected C++ encoder instrumentation and monitoring framework supporting Google Cloud, Prometheus and OpenTelemetry;
- Implemented instrumentation of FFmpeg libavfilter pipelines reducing end-to-end latency by 4 seconds;
- Developed AI skills for the team allowing faster feature delivery and incident resolution;
- Implemented end-to-end Cea608/708 to TTML (IMSC) and WebVTT subtitle conversion workflow;
- Integrated mcp-on-demand, graphify and rag-code-mcp into the AI workflows increasing response quality and reducing token consumption;
- Owned and maintained internal fork of rag-code-mcp with added support for C, C++, Java and Kotlin written in Go;
- Implemented proper encoder recovery mechanism on output outages covering S3, GCS, Azure Blob Storage and Akamai MSL4/5;
- Diagnosed and resolved HLS and DASH manifest misalignments in Live to VOD workflows and preventing orphan segments on input removal.

Environment
C++20, Python, Java, Go, Docker, Kubernetes,

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Draw.io, GitHub Actions, minikube, Grafana, Prometheus, PUML, OpenTelemetry, CircleCI, Akamai, vcpkg, CMake, ccache, SonarQube, Spring Boot, FFmpeg, libavfilter, libavformat, GTest, Spring, gRPC, Protobuf, libav, ABSL, treesitter, go-tree-sitter, MPEG-DASH, HLS, LL-HLS, LL-DASH, SRT, Zixi, Cea608, Cea708, TTML, WebVTT, MySQL, RabbitMQ, REST API, Kong, Akamai MSL4, Akamai MSL5, Microservices architecture, Claude, graphify, rag-code-mcp, mcp-on-demand, JSON RPC, Azure (SDK, Blob Storage, Functions), AWS (Lambda, EC2, S3, CloudFront), GCP (Compute Engine, Cloud Storage, Cloud Functions, Monitoring).

LIBRARY SYSTEM

A robust and highly scalable library management system with over 40 years of operational history, serving a diverse client base across the globe. The system is built using a combination of C, Java, and Scala, leveraging the strengths of each language to deliver a reliable, efficient, and feature-rich platform. Designed to handle high volumes of data and complex workflows, it has continually evolved to meet the needs of modern libraries while maintaining compatibility with legacy components.

Project roles
Software Engineer

Period
03.2022 – 04.2025

Responsibilities

- Optimized and maintained the existing codebase by resolving complex bugs and implementing performance enhancements;
- Developed new backend and frontend features for socket-based communication;
- Led development efforts as a key contributor in C language, driving the successful implementation of critical features and solutions;
- Refactored existing Scala API code to improve maintainability and scalability;
- Worked with AWS pipelines, automating CI/CD processes for smoother deployments and reducing development cycle times;
- Modified database tables to support new feature implementations, improving schema design to align with updated system workflows;
- Estimated tasks with precision, ensuring realistic timelines and efficient resource allocation for development milestones;
- Conducted thorough code reviews, ensuring code quality, adherence to standards, and knowledge

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sharing within the basic team and client's team;
- Refactored legacy application code to improve maintainability, scalability, and overall system efficiency.

Environment
C, Java 11, Scala, gdb, MySQL, PostgreSQL, Amazon Web Services, Git, BitBucket.

CORPORATE SYSTEM FOR THE OIL AND GAS INDUSTRY

Corporate system tailored for the oil and gas industry, designed for seamless internal use across various devices. The system leverages C++17 for robust backend development and JavaScript, HTML, and jQuery for interactive frontends. With Microsoft SQL Server handling data storage and Nginx ensuring smooth server operations, it offers a reliable and high-performance solution.

Development processes were streamlined with Resharper C++ for enhanced code quality, CMake for efficient build automation, and version control tools like Git and BitBucket. The solution was optimized for deployment in Windows environments, ensuring compatibility and ease of use across devices.

Project roles
Software Engineer

Period
10.2021 – 12.2021

Responsibilities
- Engineered efficient back-end software solutions, ensuring optimal performance for critical operations in the oil and gas industry;
- Contributed as a core developer, implementing key functionalities and supporting project milestones;
- Provided accurate task estimations, enabling effective planning and resource management for project deliverables;
- Maintained and enhanced the existing codebase by resolving bugs, optimizing performance, and reducing technical debt;
- Removed outdated legacy components, modernizing the system to improve reliability and maintainability.

Environment
C++17, JavaScript, HTML, Windows, CMake, Resharper C++, jQuery, MySQL, Microsoft SQL Server, Nginx, Git, BitBucket.

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HOME ACCOUNTING SOFTWARE

Bookkeeping software provides an essential solution for any organization or individual who needs to improve their degree of control over their budgeting. The software can be used to record its users transactions including bills, credit card statements, receipts and more.

Project roles
Software Engineer

Period
07.2020 – 07.2021

Responsibilities

- Developed a home accounting application to manage personal financial transactions;
- Utilized the FLTK C++ library to create a user-friendly graphical interface for simplifying transaction entry and financial analysis;
- Integrated SQLite for efficient database management of transaction records, ensuring quick access and reliable data storage;
- Implemented robust testing practices using Google Test to ensure application reliability and accuracy, resolving potential issues before deployment;
- Employed plog for effective error logging and debugging, enhancing application reliability, and managed build automation with CMake for streamlined development.

Environment
C++, CMake, Visual Studio, Gtest, FLTK, plog, SQLite, Git.

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3D RECONSTRUCTION SOFTWARE FOR INDUSTRIAL APPLICATIONS

Developed cutting-edge software aimed at generating precise 3D models by processing data captured by multiple 2D laser sensors. The solution is tailored for industries requiring highly accurate 3D spatial data, such as manufacturing, robotics, and quality control.

Project roles
Software Engineer

Period
03.2020 – 05.2020

Responsibilities

- Developed software for generating 3D models from data collected by multiple 2D laser sensors, significantly enhancing accuracy in 3D reconstruction;
- Utilized the Point Cloud Library (PCL) for efficient processing and manipulation of point cloud data, optimizing data handling and performance;
- Integrated Open3D to facilitate advanced visualization and analysis of 3D models, improving user interaction and understanding of spatial data;
- Employed Visual Studio for coding, debugging, and optimizing performance, ensuring a robust and efficient software solution.

Environment
C++, Visual Studio, PCL, Open3D, Git.

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SOFTWARE MODULES FOR UAV

Developed innovative software solutions and algorithms to enhance the performance of optical sensors used in satellites and UAVs. The project included advanced photo and video processing, data analysis, and testing to improve system reliability, accuracy, and image quality for remote sensing applications.

Project roles
Software Engineer

Period
08.2018 – 12.2019

Responsibilities

- Developed software modules for digital image processing systems used in Unmanned Aerial Vehicles;
- Implemented a debayer algorithm to convert raw sensor data into full-color images, significantly improving image quality and accuracy for better visual data interpretation;
- Created image format conversion tools to support multiple formats, enhancing compatibility and integration across different systems and platforms;
- Developed an image stabilization algorithm to reduce motion-induced distortions, improving video clarity in real-time UAV operations for clearer footage capture;
- Designed an image fusion algorithm to combine data from multiple sensors, increasing detail and contrast for improved target recognition and analysis.

Environment
MATLAB, PostgreSQL, Git, BitBucket.

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REMOTE SENSING SATELLITES DATABASE

A comprehensive database of Earth remote sensing satellites, designed to store extended data for research and operational purposes. Built using Ruby and Ruby on Rails, the system provides a robust framework for managing satellite information. Data storage and querying were handled efficiently with Microsoft Access and PostgreSQL, ensuring scalability and reliability.

This database facilitated advanced analysis and streamlined workflows for satellite data management, supporting a wide range of applications in Earth observation and remote sensing.

Project roles
Software Engineer

Period
03.2016 – 03.2017

Responsibilities

- Built a web-based interface using Ruby on Rails, enabling easy access and manipulation of satellite data for users of varying technical expertise;
- Designed and optimized a PostgreSQL database to handle large datasets with detailed satellite specifications, improving data retrieval speed and enhancing user experience;
- Integrated MS Access for legacy data support, ensuring smooth data migration and compatibility with existing systems to facilitate seamless transitions.

Environment
Ruby, Ruby on Rails, PostgreSQL, MS Access.

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SOFTWARE FOR PROCESSING AND ANALYSIS OF SATELLITE EARTH OBSERVATION IMAGES

The software includes a range of modules with functions such as determining linear resolution, enhancing image quality through prediction of image type and other factors, stitching frames taken by a remote sensing satellite, eliminating optical distortions between frames. Additionally, one of the key components of the project is the development of a compression algorithm based on Adaptive differential pulse-code modulation (ADPCM).

These features are aimed at providing a comprehensive toolkit for working with satellite imagery, enabling precise resolution analysis, effective data compression, and optimized image processing workflows for various remote sensing applications.

Project roles
Software Engineer

Period
08.2012 – 03.2016

Responsibilities

- Developed algorithms for standardizing brightness levels across multiple frames and eliminating optical defects, improving the overall quality and consistency of satellite imagery;
- Designed and implemented software to analyze the linear resolution of satellite images using the sharp edge method, enhancing the accuracy of imaging performance evaluation;
- Created algorithms to assess image sharpness and edge clarity, ensuring precise resolution measurements for remote sensing applications and contributing to reliable data interpretation;
- Developed software to predict quantization ranges for remote sensing satellites by analyzing Earth observation data and estimating the albedo range, improving sensor calibration and minimizing data loss;
- Automated the prediction of quantization ranges, streamlining workflows, and improving operational efficiency;
- Designed and implemented a custom image compression algorithm using ADPCM with multiple quantizers, achieving optimized compression ratios and enhanced flexibility for different data sizes.

Environment
C++, MATLAB, System Tool Kit, CMake, Visual Studio, PostgreSQL, MS Access, Jira, Git.

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Professional skills

SKILLS		EXPERIENCE IN YEARS	LAST USED
Programming Languages	C++	6	2026
	C	3	2025
	C#	1	2026
	Java	3	2025
	JavaScript	1	2026
	Ruby	1	2017
	MATLAB	5	2019
	Scala	3	2025
	HTML	1	2021
Operating Systems	Windows	1	2021
	Embedded Linux	1	2026
Tools	Yocto Project	1	2026
	Qt Designer	1	2026
	Qt Designer .ui forms	1	2026
	System Tool Kit	4	2016
	systemd	1	2026
	CMake	6	2026
	qmake	1	2026
	gcc	1	2026
	GDB	4	2026
	Valgrind	1	2026
	Qt Test	1	2026
	Visual Studio	5	2021
	Qt Creator	1	2026
	Resharper C++	1	2021

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SKILLS		EXPERIENCE IN YEARS	LAST USED
Frameworks & Libraries	GStreamer	1	2026
	Qt	1	2026
	Qt Widgets	1	2026
	QSS	1	2026
	Ruby on Rails	1	2017
	.NET	1	2026
	GTest	1	2021
	Node.js	1	2026
	WinForms	1	2026
	FLTK	1	2021
	plog	1	2021
	PCL	1	2020
	V4L2	1	2026
	jQuery	1	2021
	Open3D	1	2020
	Qt DBus	1	2026
Multimedia	WebRTC	1	2026
	Qt Multimedia	1	2026
	Linphone-SDK	1	2026
	liblinphone	1	2026
	mediasoup	1	2026
	libmediasoupclient	1	2026
Databases	MySQL	3	2025
	PostgreSQL	9	2025
	SQLite	2	2026
	Microsoft SQL Server	1	2021
	MS Access	5	2017

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SKILLS		EXPERIENCE IN YEARS	LAST USED
Network	SDP	1	2026
	Nginx	1	2021
	SIP	1	2026
	RTP	1	2026
	Qt WebSockets	1	2026
	SNMP	1	2026
	DHCP	1	2026
	TFTP	1	2026
Cloud	AWS	3	2025
Embedded	BitBake	1	2026
	NXP i.MX	1	2026
Development Processes	Jira	4	2016
	Asana	1	2026
Source control systems	Git	11	2026
	GitLab	1	2026
	Bitbucket	5	2025
Architecture	Qt Model/View	1	2026

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